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United States Patent Application Publication

20250286201

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

SUGIE; Kazuki et al.

ELECTRICITY STORAGE DEVICE

Abstract

An electricity storage device includes: an electricity storage module including a plurality of electricity storage cells; an adhesive material that causes the electricity storage cells to adhere to each other; and an inter-cell busbar. Each of the electricity storage cells includes a first short side surface and a second short side surface, a negative electrode terminal provided on the first short side surface, and a positive electrode terminal provided on the second short side surface. The electricity storage cells are disposed such that the positive electrode terminal and the negative electrode terminal of the electricity storage cells adjacent to each other in a Y direction are adjacent to each other. The inter-cell busbar connects the positive electrode terminal and the negative electrode terminal adjacent to each other. The adhesive material causes the first short side surface and the second short side surface to adhere to each other.

Inventors: SUGIE; Kazuki (Miyoshi-shi, JP), MORI; Akio (Toyota-shi, JP), KAMO; Yoshiro (Ashigarakami-gun, JP), NAKANO; Takahiko (Nagoya-shi, JP), SHIMONAKA; Toshihiro (Toyota-shi, JP), KOHASHI; Kenichi (Sunto-gun, JP), SODA; Yuki (Nagoya-shi, JP)

Applicant: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

Family ID: 96810286

Assignee: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

Appl. No.: 18/956319

Filed: November 22, 2024

Foreign Application Priority Data

JP	2024-033837	Mar. 06, 2024
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Publication Classification

Int. Cl.: H01M50/293 (20210101); H01M10/613 (20140101); H01M50/249 (20210101); H01M50/507 (20210101); H01M50/548 (20210101)

U.S. Cl.:

CPC H01M50/293 (20210101); H01M10/613 (20150401); H01M50/507 (20210101); H01M50/548 (20210101); H01M50/249 (20210101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-033837 filed on Mar. 6, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an electricity storage device.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication (Translation of PCT Application) No. 2022-512496 (JP 2022-512496 A) discloses a battery pack in which sub-modules each including a plurality of unit cells are coupled to each other by a busbar.

SUMMARY

[0004] In the battery pack described in JP 2022-512496 A, when an impact is applied from outside, the busbar is considered to be damaged due to occurrence of positional misalignment between the sub-modules.

[0005] The present disclosure has been made to solve the above-mentioned problem, and has an object to provide an electricity storage device capable of preventing a busbar that connects electricity storage cells to each other from being damaged.

[0006] An electricity storage device according to one aspect of the present disclosure includes: an electricity storage module including a plurality of electricity storage cells arrayed in a first direction; a case that accommodates the electricity storage module; an adhesive material that causes the electricity storage cells to adhere to each other; and an inter-cell busbar that connects, out of the electricity storage cells, electricity storage cells adjacent to each other in the first direction. Each of the electricity storage cells includes: a side surface in a second direction intersecting with the first direction; and a terminal provided on the side surface. The electricity storage cells are disposed such that the terminals of the electricity storage cells adjacent to each other in the first direction are adjacent to each other. The inter-cell busbar connects the terminals adjacent to each other. The adhesive material causes the side surfaces adjacent to each other in the first direction to adhere to each other.

[0007] In the electricity storage device according to the one aspect of the present disclosure, as described above, the adhesive material causes the side surfaces adjacent to each other in the first direction to adhere to each other. Thus, the adhesive material can prevent positional misalignment from occurring between the electricity storage cells (the side surfaces) adjacent to each other in the first direction. As a result, it is possible to prevent a stress from being applied to the inter-cell busbar connecting the terminals provided on the respective adjacent side surfaces due to the above-mentioned positional misalignment. Thus, it is possible to prevent the inter-cell busbar from being damaged.

[0008] The adhesive material may cause the adjacent side surfaces respectively provided with the adjacent terminals that are connected by the inter-cell busbar to adhere to each other. With such a

configuration, the side surfaces connected (fixed) by the inter-cell busbar can adhere to each other by the adhesive material, and hence it is possible to prevent a stress from being applied to the inter-cell busbar due to the above-mentioned positional misalignment.

[0009] The adhesive material may include a first adhesion part and a second adhesion part that are provided such that the inter-cell busbar is interposed between the first adhesion part and the second adhesion part. With such a configuration, as compared to a case in which the adhesive material includes any one of the first adhesion part and the second adhesion part, the adjacent side surfaces can be more stably (firmly) fixed to each other. As a result, it is possible to prevent the inter-cell busbar from being damaged.

[0010] The electricity storage device may include a cooler including a cooling surface on which the electricity storage cells are arrayed. The adhesive material may be provided on the cooler side with respect to the inter-cell busbar. With such a configuration, it is possible to easily prevent positional misalignment from occurring between the side surfaces on the cooler side with respect to the inter-cell busbar. Thus, the present disclosure is particularly effective when the above-mentioned positional misalignment is liable to occur at a position on the cooler side due to a relatively weak adhesive force between the electricity storage cell and the cooler.

[0011] The adhesive material may have a band shape extending in the first direction. With such a configuration, the adhesion area between the adhesive material and the side surface can be easily increased. As a result, the adhesive force by the adhesive material can be easily increased.

[0012] According to the present disclosure, it is possible to prevent the busbar connecting the electricity storage cells to each other from being damaged.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0014] FIG. 1 is a view illustrating a configuration of a vehicle on which an electricity storage device according to one embodiment is mounted;

[0015] FIG. 2 is an exploded perspective view illustrating a configuration of the electricity storage device according to the one embodiment and a vehicle framework;

[0016] FIG. 3 is an exploded perspective view illustrating a detailed configuration of the electricity storage device according to the one embodiment;

[0017] FIG. 4 is a perspective view illustrating a configuration of the electricity storage device according to the one embodiment;

[0018] FIG. 5 is a sectional view illustrating the configuration of the electricity storage device according to the one embodiment; and

[0019] FIG. 6 is a plan view illustrating the configuration of the electricity storage device according to the one embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

[0020] Hereinafter, with reference to the drawings, embodiments and modification examples according to the present disclosure are described. In the following description, like parts and components are denoted by like reference symbols. The same holds true for the names and the functions thereof. Thus, detailed description of those is not repeated. It is to be noted that the embodiments and the modification examples described below may be selectively combined as appropriate.

[0021] With reference to FIG. 1 to FIG. 6, an electricity storage device according to this embodiment and a vehicle including the electricity storage device are described. FIG. 1 is a side

view schematically illustrating a vehicle **200** including an electricity storage device **100** according to this embodiment. It is to be noted that an X direction, a Y direction, and a Z direction as used herein are directions orthogonal to each other. For example, the X direction and the Y direction are a front-rear direction and a width direction, respectively, of the vehicle **200** when the electricity storage device **100** is mounted on the vehicle **200**. Further, the Z direction is the vertical direction. It is to be noted that the X direction and the Y direction are examples of a “second direction” and a “first direction” of the present disclosure, respectively.

[0022] With reference to FIG. **1**, the electricity storage device **100** is disposed below a floor panel **213** (FIG. **2**) of the vehicle **200**. Examples of the vehicle **200** include a hybrid electric vehicle, a plug-in hybrid electric vehicle, a fuel cell electric vehicle, and a battery electric vehicle. The vehicle **200** includes the electricity storage device **100** and a vehicle framework **210**.

[0023] FIG. **2** is an exploded perspective view schematically illustrating the electricity storage device **100** and the vehicle framework **210**. With reference to FIG. **2**, the vehicle framework **210** includes a left frame **211** and a right frame **212**.

[0024] The left frame **211** and the right frame **212** are disposed at a bottom portion of the vehicle framework **210**. The left frame **211** and the right frame **212** are disposed with an interval in the width direction (the Y direction) of the vehicle **200**. Further, each of the left frame **211** and the right frame **212** is disposed so as to extend in the front-rear direction (the X direction) of the vehicle **200**.

[0025] The floor panel **213** is provided between the left frame **211** and the right frame **212**. The electricity storage device **100** is disposed below the floor panel **213**, and is fixed to the left frame **211** and the right frame **212**.

[0026] FIG. **3** is a perspective view schematically illustrating the electricity storage device **100**. With reference to FIG. **3**, the electricity storage device **100** includes an electricity storage module **1** including a plurality of electricity storage cells **10**, and a case **20** that accommodates the electricity storage module **1**.

[0027] The electricity storage cell **10** is a secondary battery, and is typically a lithium-ion secondary battery. The lithium-ion secondary battery is a battery using lithium as a charge carrier, and may include, in addition to a lithium-ion secondary battery using a liquid electrolyte, an all-solid-state battery using a solid electrolyte. It is to be noted that the electricity storage cell **10** is not limited to the lithium-ion secondary battery, and may be configured of a nickel-hydrogen secondary battery or other secondary batteries.

[0028] Each of the electricity storage cells **10** is disposed so as to extend in the front-rear direction (the X direction) of the vehicle **200** (see FIG. **1**). Further, the electricity storage cells **10** are arrayed in the width direction (the Y direction) of the vehicle **200**.

[0029] The case **20** includes an upper cover **21** and a lower case **22**. FIG. **3** illustrates the electricity storage device **100** in a state in which the upper cover **21** is detached. The lower case **22** includes a bottom plate **22a**, a peripheral wall **22b**, and a plurality of partition walls **22c**, **22d**, **22e**, **22f**, **22g**. The partition walls **22c**, **22d**, **22e**, **22f**, **22g** divide a space in the case **20** into a plurality of sections.

[0030] The bottom plate **22a** is formed in a flat plate shape. The peripheral wall **22b** is formed so as to extend upward in the vehicle **200** from an outer peripheral edge portion of the bottom plate **22a**. The peripheral wall **22b** is formed in an annular shape. The partition walls **22c**, **22d**, **22e**, **22f**, **22g** are provided on the bottom plate **22a**. The partition walls **22c**, **22d**, **22e**, **22f** are each formed so as to extend in the front-rear direction (the X direction) of the vehicle **200**. The partition wall **22g** is formed so as to extend in the width direction (the Y direction) of the vehicle **200**.

[0031] The partition wall **22g** is disposed at the center in the front-rear direction (the X direction) of the vehicle **200**. Each of the partition wall **22c** and the partition wall **22f** is disposed at the center in the width direction (the Y direction) of the vehicle **200**. The partition wall **22c** is disposed on the front side (the X1 side) of the vehicle **200** than the partition wall **22g**. The partition wall **22f** is disposed on the rear side (the X2 side) of the vehicle **200** than the partition wall **22g**. A plurality of electricity storage cells **10** is accommodated in each space divided by the partition wall **22c**, the

partition wall **22d**, the partition wall **22e**, the partition wall **22f**, and the partition wall **22g**.

[0032] The lower case **22** further includes a plurality of support portions **22h** and a plurality of support portions **22i**. The support portions **22h** and the support portions **22i** are each fixed to the vehicle framework **210** (see FIG. 2). For example, the support portions **22h** and the support portions **22i** each have holes into which bolts are fitted. When the bolts are fitted into the holes, each of the support portions **22h** is fixed to the left frame **211** (see FIG. 2), and each of the support portions **22i** is fixed to the right frame **212** (see FIG. 2).

[0033] FIG. 4 illustrates an example of the electricity storage cell **10**. With reference to FIG. 4, the electricity storage cell **10** includes an upper surface **11**, a lower surface **12**, a short side surface **13**, a short side surface **14**, a long side surface **15**, and a long side surface **16**. It is to be noted that each of the short side surface **13** and the short side surface **14** is an example of a “side surface” of the present disclosure.

[0034] Each of the upper surface **11** and the lower surface **12** is a surface of the electricity storage cell **10** in the Z direction. Specifically, the upper surface **11** is an end surface of the electricity storage cell **10** on the Z1 side. The lower surface **12** is an end surface of the electricity storage cell **10** on the Z2 side, and is a surface provided on the opposite side from the upper surface **11** in the Z direction.

[0035] Each of the short side surface **13** and the short side surface **14** is a surface of the electricity storage cell **10** in the X direction. Specifically, the short side surface **13** and the short side surface **14** are one end surface of the electricity storage cell **10** in the X direction and the other end surface of the electricity storage cell **10** in the X direction, respectively.

[0036] Each of the long side surface **15** and the long side surface **16** is a surface of the electricity storage cell **10** in the Y direction. Specifically, the long side surface **15** and the long side surface **16** are one end surface of the electricity storage cell **10** in the Y direction and the other end surface of the electricity storage cell **10** in the Y direction, respectively.

[0037] The electricity storage cell **10** is configured to have a longitudinal direction in the X direction. Specifically, a width W1 of the electricity storage cell **10** in the X direction is larger than a width W2 of the electricity storage cell **10** in the Y direction. Further, the width W1 is larger than a height H of the electricity storage cell **10** in the Z direction. It is to be noted that the height H is larger than the width W2.

[0038] The electricity storage cell **10** further includes a positive electrode terminal **17** and a negative electrode terminal **18**. The positive electrode terminal **17** is provided on the short side surface **14**. The negative electrode terminal **18** is provided on the short side surface **13**. It is to be noted that each of the positive electrode terminal **17** and the negative electrode terminal **18** is an example of a “terminal” of the present disclosure.

[0039] FIG. 5 is a sectional view schematically illustrating the floor panel **213** and the electricity storage device **100**. Further, FIG. 5 illustrates the short side surface **13** and the short side surface **14** as viewed from the X1 side.

[0040] With reference to FIG. 5, in the electricity storage device **100**, the electricity storage cell **10** disposed such that the short side surface **13** faces the front (the X1 side) of the vehicle **200** (see FIG. 1) and the electricity storage cell **10** disposed such that the short side surface **14** faces the front of the vehicle **200** are alternately disposed along the width direction (the Y direction) of the vehicle **200**. Thus, the positive electrode terminal **17** and the negative electrode terminal **18** of the electricity storage cells **10** adjacent to each other in the Y direction are disposed so as to be adjacent to each other.

[0041] The electricity storage device **100** includes an adhesive material **30** that causes the electricity storage cell **10** to adhere to the upper cover **21**. The adhesive material **30** is made of resin. The adhesive material **30** is provided on the upper surface **11** of the electricity storage cell **10**. More specifically, the adhesive material **30** is provided between the upper surface **11** of the electricity storage cell **10** and the upper cover **21**, and along the upper surface **11**. The electricity

storage cells **10** are fixed to the upper cover **21** by the adhesive material **30**.

[0042] The electricity storage device **100** further includes a cooler **40** that cools the electricity storage cell **10**. The cooler **40** includes a cooling surface **41** on which the electricity storage cells **10** are disposed. The cooling surface **41** is an end surface of the cooler **40** on the **Z1** side. The cooler **40** (cooling surface **41**) is provided along the lower surface **12** of the electricity storage cell **10**. It is to be noted that FIG. **5** illustrates an example in which no adhesive material (adhesive layer) is provided between the cooling surface **41** and the lower surface **12** of the electricity storage cell **10**, but the adhesive material (adhesive layer) may be disposed at this position.

[0043] The electricity storage device **100** further includes an insulating plate **50**. The insulating plate **50** is provided between the cooler **40** and the bottom plate **22a** and along the bottom plate **22a**.

[0044] The electricity storage device **100** includes a plurality of inter-cell busbars **60**. The inter-cell busbar **60** connects the electricity storage cells **10** adjacent to each other in the **Y** direction. Specifically, the inter-cell busbar **60** connects the positive electrode terminal **17** and the negative electrode terminal **18** adjacent to each other. In detail, the inter-cell busbar **60** connects the positive electrode terminal **17** provided on one of the two electricity storage cells **10** adjacent to each other in the **Y** direction and the negative electrode terminal **18** provided on the other of the two electricity storage cells **10**.

[0045] It is to be noted that, as illustrated in FIG. **6**, also on the **X2** side of the electricity storage module **1** opposite to the **X1** side illustrated in FIG. **5**, similarly, the inter-cell busbar **60** connects the positive electrode terminal **17** and the negative electrode terminal **18** adjacent to each other. Each electricity storage cell **10** (an electricity storage cell **10** other than the electricity storage cells **10** at both ends in the **Y** direction) is connected on the **X1** side to the electricity storage cell **10** disposed on one side in the **Y** direction by the inter-cell busbar **60**, and is connected on the **X2** side to the electricity storage cell **10** disposed on the other side in the **Y** direction by the inter-cell busbar **60**. Thus, the electricity storage cells **10** arrayed in the **Y** direction are electrically connected in series.

[0046] Here, in the electricity storage device of the related art, it is considered that the inter-cell busbar is damaged due to occurrence of positional misalignment between the adjacent electricity storage cells **10** when an impact is applied from outside or the like.

[0047] In view of the foregoing, in this embodiment, with reference back to FIG. **5**, the electricity storage device **100** includes an adhesive material **70** that causes the short side surface **13** and the short side surface **14** adjacent to each other in the **Y** direction to adhere to each other. Specifically, the adhesive material **70** causes the short side surface **14** and the short side surface **13** provided with the positive electrode terminal **17** and the negative electrode terminal **18** that are connected by the inter-cell busbar **60**, respectively, to adhere to each other. Accordingly, the combination of the electricity storage cells **10** that are caused to adhere by the adhesive material **70** is different between the **X1** side and the **X2** side of the electricity storage module **1** (see FIG. **6**).

[0048] Further, as illustrated in FIG. **5**, the adhesive material **70** includes a first adhesion part **71** and a second adhesion part **72**. Each inter-cell busbar **60** is provided so as to be interposed between the first adhesion part **71** and the second adhesion part **72**. Specifically, the first adhesion part **71** is provided on the **Z1** side from the inter-cell busbar **60**. The second adhesion part **72** is provided on the **Z2** side from the inter-cell busbar **60**. It is to be noted that the first adhesion part **71** and the second adhesion part **72** have the same shape and the same size.

[0049] The second adhesion part **72** is provided on the cooler **40** side with respect to the inter-cell busbar **60**. In other words, the second adhesion part **72** is disposed between the inter-cell busbar **60** and the cooler **40**.

[0050] Further, each inter-cell busbar **60** is disposed below (on the **Z2** side of) the center of the electricity storage cell **10** in the **Z** direction. In contrast, the first adhesion part **71** is provided in the vicinity of the center of the electricity storage cell **10** in the **Z** direction. Further, the second

adhesion part **72** is provided in the vicinity of the lower surface **12** of the electricity storage cell **10**.
[0051] The adhesive material **70** (**71**, **72**) has a band shape extending in the Y direction. The adhesive material **70** has a width **W11** in the Y direction. The adhesive material **70** has a width **W12** in the Z direction. The width **W11** is larger than the width **W12**. The width **W11** is, for example, four times the width **W12** or more.
[0052] The width **W11** of the adhesive material **70** in the Y direction is, for example, larger than the width **W2** of the electricity storage cell **10** in the Y direction (FIG. **4**). Further, the width **W11** is larger than a distance **D** in the Y direction between the positive electrode terminal **17** and the negative electrode terminal **18** that are connected by the inter-cell busbar **60**.
[0053] As described above, in the above-mentioned embodiment, the adhesive material **70** causes the short side surface **13** and the short side surface **14** adjacent to each other in the Y direction to adhere to each other. Thus, the adjacent electricity storage cells **10** are fixed to each other by the adhesive material **70**, and hence it is possible to prevent a stress from being applied to the inter-cell busbar **60** connecting the adjacent electricity storage cells **10**. As a result, it is possible to prevent the inter-cell busbar **60** from being damaged.
[0054] Further, in the above-mentioned embodiment, the second adhesion part **72** is provided on the cooler **40** side with respect to the inter-cell busbar **60**. Thus, even when each electricity storage cell **10** is not caused to adhere to the cooler **40**, the second adhesion part **72** can prevent the positional misalignment from occurring between parts of the adjacent electricity storage cells **10** on the cooler **40** side.
[0055] The above-mentioned embodiment has shown an example in which the negative electrode terminal **18** is provided on the short side surface **13** of the electricity storage cell **10**, and the positive electrode terminal **17** is provided on the short side surface **14** of the electricity storage cell **10**, but the present disclosure is not limited thereto. The positive electrode terminal **17** and the negative electrode terminal **18** may be provided on any one of the short side surface **13** and the short side surface **14**. In this case, the adjacent short side surfaces (**13** or **14**) provided with the terminals (**17**, **18**) are caused to adhere to each other by the adhesive material. It is to be noted that the adjacent short side surfaces (**13** or **14**) not provided with the terminals (**17**, **18**) may be caused to adhere to each other by the adhesive material.
[0056] The above-mentioned embodiment has shown an example in which the first adhesion part **71** and the second adhesion part **72** are used for adhesion between the adjacent electricity storage cells **10**, but the present disclosure is not limited thereto. Any one of the first adhesion part **71** and the second adhesion part **72** may be used for adhesion between the adjacent electricity storage cells **10**.
[0057] The above-mentioned embodiment has shown an example in which the adhesive material **70** causes the side surfaces (**13**, **14**) provided with the terminals (**17**, **18**) that are connected by the inter-cell busbar **60** to adhere to each other, but the present disclosure is not limited thereto. For example, the adhesive material may cause the adjacent side surfaces (**13**, **14**) provided with the terminals (**17**, **18**) that are not connected by the inter-cell busbar **60** to adhere to each other.
[0058] The above-mentioned embodiment has shown an example in which each adhesive material **70** adheres the two adjacent electricity storage cells **10** to each other, but the present disclosure is not limited thereto. The adhesive material may cause three or more adjacent electricity storage cells **10** to adhere to each other.
[0059] The above-mentioned embodiment has shown an example in which the first adhesion part **71** and the second adhesion part **72** have the same shape and the same size, but the present disclosure is not limited thereto. The first adhesion part and the second adhesion part may have different shapes and different sizes. For examples, the width of the second adhesion part in the Y direction may be larger than the width of the first adhesion part in the Y direction. Further, the thickness of the second adhesion part in the X direction may be larger than the thickness of the first adhesion part in the X direction.

[0060] The above-mentioned embodiment has shown an example in which the cooler **40** is provided below the electricity storage cell **10**, but the present disclosure is not limited thereto. The cooler may be provided above the electricity storage cell.

[0061] The above-mentioned embodiment has shown an example in which the electricity storage cells **10** are arrayed in the Y direction orthogonal to (intersecting with) the Z direction (height direction), but the present disclosure is not limited thereto. For example, the electricity storage cells **10** may be arrayed (stacked) in the Z direction.

[0062] The above-mentioned embodiment has shown an example in which the electricity storage device **100** is mounted on the vehicle **200**, but the present disclosure is not limited thereto. The electricity storage device **100** may be provided on an electric device (for example, a stationary electricity storage device) other than the vehicle.

[0063] The above-mentioned embodiment has shown an example in which the first adhesion part **71** is provided at the center of the electricity storage cell **10** in the Z direction, but the present disclosure is not limited thereto. For example, the first adhesion part **71** may be provided in an upper portion of the electricity storage cell **10** (for example, in the vicinity of the upper surface **11**).

[0064] The above-mentioned embodiment has shown an example in which the adhesive material **70** (**71**, **72**) extends in the Y direction, but the present disclosure is not limited thereto. The adhesive material **70** (**71**, **72**) may extend in the Z direction. Further, the adhesive material **70** (**71**, **72**) may extend so as to intersect with each of the Z direction and the Y direction.

[0065] The above-mentioned embodiment has shown an example in which the adhesive material **70** (**71**, **72**) has a band shape, but the present disclosure is not limited thereto. The adhesive material **70** (**71**, **72**) may have a shape other than the band shape (for example, a square shape and a circular shape).

[0066] It should be understood that the embodiments disclosed herein are illustrative and non-restrictive in every respect. The scope of the present disclosure is defined by the terms of the claims, rather than the description above, and is intended to include any modifications within the scope and meaning equivalent to the terms of the claims.

Claims

1. An electricity storage device comprising: an electricity storage module including a plurality of electricity storage cells arrayed in a first direction; a case that accommodates the electricity storage module; an adhesive material that causes the electricity storage cells to adhere to each other; and an inter-cell busbar that connects, out of the electricity storage cells, electricity storage cells adjacent to each other in the first direction, wherein each of the electricity storage cells includes: a side surface in a second direction intersecting with the first direction; and a terminal provided on the side surface, the electricity storage cells are disposed such that the terminals of the electricity storage cells adjacent to each other in the first direction are adjacent to each other, the inter-cell busbar connects the terminals adjacent to each other, and the adhesive material causes the side surfaces adjacent to each other in the first direction to adhere to each other.
2. The electricity storage device according to claim 1, wherein the adhesive material causes the adjacent side surfaces respectively provided with the adjacent terminals that are connected by the inter-cell busbar to adhere to each other.
3. The electricity storage device according to claim 1, wherein the adhesive material includes a first adhesion part and a second adhesion part that are provided such that the inter-cell busbar is interposed between the first adhesion part and the second adhesion part.
4. The electricity storage device according to claim 1, further comprising a cooler including a cooling surface on which the electricity storage cells are arrayed, wherein the adhesive material is provided on the cooler side with respect to the inter-cell busbar.

5. The electricity storage device according to claim 1, wherein the adhesive material has a band shape extending in the first direction.
